

BUSINESS PLAN REPORT

SCOPE	MW JAPAN PIC	SECTION	ISSUES AT MM	IMPLEMENTATIONS AT MM	IMPLEMENTATION AT JM	Remark 1	Remark 2	Remark 3
MILL EQUIPMENT	DAIMON	BALLMILL	1) Manual control of ballmill operation for running time etc --> Analog and digital timers used --> Time set for first and second process but can be changed by anyone --> Running time change usually if slurry does not meet QC spec (particle size)	1) Only ballmill temperature recorded in system with alerting	1) How does the system work, what aspects are controlled/monitored?	System specificationn and design taken. Currenty used as control in JM. For MM side need to implement this system and extend on IOT to use data for PM (CMMS), Abnormality etc	-	-
WASTEWATER TREATMENT & FACILITY	KOJIMA	WWTP	1) Items stored at the plant	1) tools, pumps, polymer chemical, PPE	1) Site visit for 7s evaluation	No system with sensors to control currently excpt in east wing block theres a plan to do it further. recycling activities are carried out upto 80% of capacity. RO system is installed. Around 4m3/hr of water is passing through the RO filter.] running costs of the RO is about Yen160,000 / mth. The filter change is done 2 years once. The recycle water is used back into process. ** as explained above, only in the east wing	-	MM should consider the RO since the equipment is partially availableand recycly the filtered water
			2) Sensors implemented	2) pH, flow and turbidity	1) Are there any additional sensors in addition to pH, flow and turbidity? iOT capable? Does it assist in ensuring compliant water only is released from the facility?			
			3) Outflow countermeasures	3) Semi auto valve and pump back system	1) Are there any measures for non-compliant wastewater to be sent back to the plant for re-treatment?			
			4) No mechanism to prevent tampering of chemical dosing rate	-	1) Can the Mw Japan ballmill system be used to control and lock the dosing qty at the decided value?			
			5) Recycling system for WWTP 3 to reuse for flushing and chiller	5) Building 2 layer filtration to ensure sustained good water quality --> Process filter convert from silica sand to zeolite --> Utilise 3 units sand, carbon and softener filter previously used for RO	1) Are there any planned activities for treated water to be recycled for reuse? -Malaysia plans to reuse water for chiller and toilet flushing. Does Toki plant machinery chillers utilise any additives in the circulating chilled water? (eg glycol)			
FIRE PREVENTION EQUIPMENT	KUWASHITA	FIRE SYSTEM & EMERGENCY RESPONSE		1) MFAP for each plant, sprinkler, FE, Foam (Chemical Tank), heat and smoke detection, breakglass, hydrant, hose reel	1) How does the fire protection system in Mw Japan work? What are the components within the fire protection system?	Same equipment but no foam required. Not a practice in Japan.	-	Malaysia can structure under same concept
				2) ERT organisation and hierarchy of notification & guard house	1) Is there an emergency response team? How is the communication carried out in the event of emergencies?	ERT training once a year by outside contractor. Limited walkie talkie is used and main reference point is to security like in Malaysia. Their ERT orgaanisation is made into 1) Fire Protection Group; 2) Evacuation guidance group; 3) communication group; 4) Public Relations Group; 5) Record Keeping Group; 6) Items Recovery Group		

ESG MANAGEMENT METHOD	KAWAJIRI(MS.) INAGAKI	ESG	Malaysia Scope - WWTP (water recycling), firing (gas and electricity usage), Renewable Energy		1) What aspects is Mw Japan looking at for further emissions reduction?	in Japan Scope is decided by Government and followed by Group . For Scope 1 is mainly Energy Consumption which is derived by provider website in total. For individual factory/units the sensor is placed at incoming pannel and connected to a trans receiver to get information of consumption. For Scope 2 is manual derived based on incoming purchase data. and Vehicle is only company vehicle. For Scope 3 which on waste generation, JM has 7 types of major waste which is reclyes, landfill waste and other waste related to plating which is mixed with some substance then fired to evaporate	currently tedious excel sheet maintained. Propose both MM and JM with single format to be used by group	Under New system to design for JM also on waste management
SAFETY, HEALTH & ENVIRONMENT RISKS @ MM		LAYOUT SYSTEM (GEMBA)			1) Are there any updates on the implementation of the layout system? Has further information been received on the usability/capability of the system?			
		CASTING	1) PPE for sharp parts cleaning/adjustment --> Input - Doctor Blade --> Output - Slitter blade	1) Current implementation --> cut resistant arm guard and double layer kevlar weave glove (glove require frequent changes) --> Equipment - Replace L Allen key with T allen key for better control and grip	1) What PPE & tools are being used for jobs in this area? Is there significant injury risk?	Sample obtained from Japan. Already in use and proven. Will check for MM on same grade cut proof glove	-	-
			2) Output green sheet cutter blade employee access prevention	2) Cover is sealed using screw requiring specialised equipment to open	1) Are there any deployed methods to prevent access into the area of the blade's movement?	Phot taken on Japan model. To obtain drawing from japan		
			3) Panel window sealing gasket for smell		1) What kind of sealing gasket is used for the panel window? Can sample be brought back to test retrofit one machine?	Seal spec to be obtain from Nakano san.		
			4) Coil handling	4) Done manually by opt from mc to jack and jack to trolley	1) Are there any methods to reduce the manual handling of green sheet coils from machine to storage trolley?	-		
		VACUUM TANK	1) Cleaning process (How is tank lid remove	1) Use chain block and manual lock once cap raised sufficiently high --> Plan to utilise winch with locking mechanism in the event winch fails and cap falls before manual locking height can be achieved	1) How is the tank lid lifted for cleaning to be done and are there measures to prevent the cap from dropping?			
		CHEMICALS	1) Warehouse storage	1) No clear segregation of materials besides by packaging size	1) How are the chemicals segregated and stored? How is the break bulk process done? If additional containers are used to break bulk, what kind of containers are they?	-Up to individual plant but safety will audit to check for compliance - Label bottles are used - Outside part is auditing something like 3 monrths once etc beside safety group -no legal rules and safety does not limit. N / R		
			2) Break bulk qty storage in prod	2) Stored on pallets/racks, using carboys and plastic containers				
			3) Containers for daily chemical use	3) Utilise plastic carboys with label				
			4) Labelling	4) Short SDS, chemical name and container label	1) How is chemical labelling practiced?			
			5) Safe Inventory Limit	-	1) Is there a guideline for the safe limit for storage? (Production, Warehouse, RnD)			
			6) Outflow/Spillage Mitigation	6) Drain with sump pit, containment units	1) What implementations are practiced in Japan to prevent spillages and to contain them if they happen? Is there a method to determine qty of absorbent to be stored at any given location? -How are chemical drums stored where breaking bulk into smaller containers is required?			

		CHEMICAL UNDERGROUND TANK	1) insufficient containment if failure occurs at tanker	1) utilise only spill kit. Malaysia does not have guidelines on qty of spill kit to be stored	1) Are there implementations to contain spillages from tankers/pipelines and prevent them from entering drains?	Old Facility (Manual)		
		UNIFORM	1) Uniform for ERT members need to stand out	1) All operators wear the same designated uniform.	Does Japan implement separate uniforms for those affiliated with emergency response or ESH in general?	No special uniform in JM. To be jointly decided for mutual implementation		
		PPE STORAGE	1) No storage provided at the factory	1) All PPE is brought by employees when they come to work and taken home after work.	1) method deployed by Japan for PPE storage on site. Has it been standardised? What method has been deployed to prevent misuse of storage locations?	Locker is provided but not only for PPE. Replacement of ahoe etc is generally set around 1 year lifespan	MM need to impose	
		QUALIFIED PERSONNEL		1) SAFETY - 2 (OHC)	1) Hows the ratio decided in JM? Any regulation or	Numbers are based on law requirement. fire prevention, hazardous material, high pressure gas, electrical, septic tank, industrial waste	To obtain JM organisation and benchmark with Malaysia. MM need to have a system to register qualified personnel	there might be a future plant to expand also might use experience senior production as a ESH member
				2) ENVIRONMENT - Scheduled Waste (1), WWTP (1)				
				3) CONFINED SPACE --> PMM SUPPLIER (1)				
	Cafeteria		1) New implementation as per in JM		1) MM does now have much facility other then vending mc etc	1) Need to introduce toast mc, digital card for payment	2) Centralise communication system for canteen display	

7s			RACKING SYSTEM OLD AND NOT SYSTEMIZE		1) How the Maintenance dept store their spare parts or tools in production.	1) in JM too its not properly standardised. A best prectice needs to be established		
			PPE SAFEKEEPING WATER BOTTLE AREA		1)How employee keep their PPE when going to break time/washroom 2) Where employee keep the water bottle in production floor.	Theres a water bottle storage cabinet/jib designed (As per photo)		

SN - Powder Processing

In addition to the above, the powder preparation process was inspected at SN. Powder mixing done using 2 chemical types and 2 process involve (vibrator) to break the particle. Next process is powder is compressed in 2 mould size (1/4 and 1) and placed in for sintering max degree 1430c.

The hardened output is then goes for crushing primary and secondary process then into attraitor and echng process for improve thermal conductivity. Spray dryer then applied for final powder preparation.

Safety Concerns
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- 1) Crushing process handling
- 2) HF Chemical handling while concentrated form